

# DEEPAK KHIMAVATH

AI Platform Engineer · Developer Tools · Agentic Systems

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Bengaluru, India

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## AI platform and developer tools engineer who builds production systems, not demos.

Designed, built, and shipped a multi-provider LLM routing Visual Studio extension, a serverless PR review agent (**150+ pull requests** reviewed org-wide in 30 days), and a **6-agent** code pipeline backed by a dual-layer RAG system — each as the sole engineer. Resolved a **15–20 hour processing bottleneck down to 15 minutes** across a distributed event-driven architecture with **50+** microservices. Recognized by the Director of Engineering and Chief Solution Architect for driving the organization's AI platform and agent engineering direction.

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## CORE COMPETENCIES

**AI & Systems:** Multi-agent systems · LLM infrastructure · GenAI / RAG · retrieval pipelines · vector databases (Qdrant) · prompt engineering & evaluation · AI observability (LangFuse) · tool calling · AI copilots · MCP (Model Context Protocol) · workflow automation · code intelligence · AI platform engineering

**Languages:** Python · C# · JavaScript · Java · C/C++ · SQL

**Cloud & DevOps:** Azure Functions · Azure AI (Claude, GPT-5) · Azure DevOps REST API · AWS · DigitalOcean · Docker · GitHub Actions · VSIX / MSBuild

**Backend & Infra:** FastAPI · Flask · Node.js · event-driven architecture · distributed microservices · message queues · WebView2 · MSAL.NET · OAuth 2.0 · JWT

**Tooling:** Visual Studio SDK (WPF, VSCT) · VS Code extension API · LlamaIndex · Qdrant · sentence-transformers · PyTorch · TensorFlow · REST APIs · JSON processing · Power BI

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## EXPERIENCE

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### Trainee Engineer · Eton Solutions

Jul 2025 – Present

Wealth management and Financial services platform · 50+ microservice event-driven architecture · AI tooling · Recognized by Director of Engineering & Chief Solution Architect

- ▶ **EDA Throughput:** Diagnosed and resolved a routing bottleneck in the core message-queue layer — cutting end-to-end processing latency from **15–20 hours to under 15 minutes** across a **50+ service distributed system**, with no regression on dependent services.
- ▶ **Microservice Ownership:** End-to-end ownership of three production financial microservices: EliminationService (trade elimination), DFRulesProcessorService (rules evaluation engine), and JournalEntryPersistService (ledger persistence) — design through deployment.
- ▶ **AI Platform:** Designed and delivered the engineering team's internal AI infrastructure — a VS extension, an automated PR reviewer, and a multi-agent code pipeline with distributed AI workflows and AI execution pipelines. Worked directly with the Director of Engineering and Chief Solution Architect on architecture decisions; all three systems are in active production use.

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### Full-Stack Developer Intern · SpurzeeTechnologies

Jul 2024 – Jun 2025

Algorithmic trading systems · LLM signal integration · AWS / DigitalOcean

- ▶ **Trading Platform:** Built a real-time stock analysis system with automated options Buy/Sell execution, LLM-assisted trade signal generation, and market pattern detection — improving data processing throughput by **25%**.
- ▶ **Price Forecasting:** Trained and deployed a Random Forest regression model for stock price prediction; managed full lifecycle from model development through cloud production deployment.

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### Alumni Mentor — Agentic AI & Developer Tooling · PES Institute of Technology and Management

Invited back to campus · Guest instruction for junior engineers

- ▶ Conducted sessions on practical AI platform engineering — multi-agent system design, GenAI / RAG architecture, retrieval pipelines, AI observability, and developer tooling — focused on the gap between academic ML and production AI work.
- ▶ Mentored students on building real systems: agent orchestration patterns, prompt engineering discipline, and how AI-assisted developer workflows are structured in industry.

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## ENGINEERING PROJECTS

## PR Review Agent — Serverless Code Review Platform — Python · Azure Functions · Azure DevOps · Mistral / Claude / GPT-5 · LangFuse

Production · Org-wide · 150+ PRs reviewed in 30 days

- ▶ **Architecture:** Dual-function serverless design — HTTP trigger acknowledges the Azure DevOps webhook in under 1 second; review work runs in a queue-triggered background processor with a 9-minute budget, handling full diff analysis, Jira context retrieval, and LLM evaluation without blocking the DevOps pipeline.
- ▶ **Delta reviews:** On each new commit, the agent reads its own prior thread comments, extracts previous findings, and instructs the model to assess whether each issue was fixed, partially addressed, or remains — avoiding redundant full re-reviews.
- ▶ **Developer interface:** Conversational @agent command system inside PR threads: **re-review**, skip, focus, context inject, explain. Handles 4 Azure DevOps webhook payload variants for thread ID resolution, with REST API fallback for thread matching.
- ▶ **Reliability:** 900-line system prompt across 7 review dimensions with structured prompt evaluation; confidence-based finding downgrade (MEDIUM/LOW blocking → warning); AI observability via LangFuse for tracing and monitoring LLM calls; commit-keyed deduplication ensures the same code is never reviewed twice.

## Eton Dev — AI Development Companion (Visual Studio and VS Code Extension) — C# · .NET · WebView2 · Python · MSAL.NET · Azure DevOps · Jira

Production · Internal engineering platform · Sole architect

- ▶ **Developer workflow:** VSIX extension for Visual Studio 2022/2026 with a single WebView2 panel covering AI chat, pre-PR code review, bug investigation, one-click code fixes, and Azure DevOps PR creation using local Git command infrastructure — consolidating what previously required switching between 5+ tools.
- ▶ **LLM routing:** Protocol-based provider abstraction supporting Claude (Azure AI), Mistral, GPT-5 (Azure OpenAI), and a custom local backend — with repo-aware routing rules and retry logic on transient failures.
- ▶ **Auto-fix engine:** Two-phase pipeline: a planner validates whether a finding is safely auto-fixable and bounds the edit to  $\pm 3$  lines; a patch generator produces anchored old/new block replacements with atomic writes and automatic backup before modification.
- ▶ **Auth:** Silent MSAL WAM broker authentication with AAD tenant auto-discovery; browser-based SSO fallback for Conditional Access policies; in-memory JWT caching with expiry verification — built for a managed corporate environment.

## Eton ARC — 6-Agent Code Intelligence Pipeline — Python · Claude Opus 4.5 / Sonnet / Haiku · Qdrant · LlamaIndex · FastAPI

Production prototype · Tested on live tickets · 11 services in index

- ▶ **Pipeline:** Triage → grep-first Discovery → parallel Specialist Council (one agent per affected service) → cross-service Moderator → Opus principal review → Execute. Takes a Jira ticket as input; produces verified file-level diffs, a PR description, and xUnit test stubs — structured for handoff to Claude Code or Codex.
- ▶ **Code retrieval:** Dual-layer GenAI retrieval pipeline built on Llama Index and Qdrant: source code chunked at 80 lines with 15-line overlap (384-dim embeddings, 11 services), plus structured service profiles in 3 formats. Exact identifier grep handles deterministic context retrieval; vector search activates when identifier matching is insufficient.
- ▶ **Review gate:** Opus issues a GO / CONDITIONAL-GO / NO-GO verdict with per-file evidence, acceptance criteria coverage map, regression risk notes, and a clean list of removed scope. Developer reviews the plan before any code is written or applied.

## LSTM + Sentiment Analysis — Stock Price Forecasting — Python · PyTorch · LSTM · NLP · scikit-learn

Research · 15% accuracy improvement · Benchmarked against 5 models

- ▶ Combined LSTM with financial news sentiment signals; benchmarked against Linear Regression, Random Forest, K-NN, and ANN. **15%** improvement in forecast accuracy. Full preprocessing, training, and evaluation pipeline.

## EDUCATION

### B.E. Computer Science & Engineering

PES Institute of Technology and Management, Shivamogga · GPA: 8.62 / 10

2021 – 2025

## RECOGNITION

- ▶ **Director of Engineering & Chief Solution Architect** — Recognized for building and delivering the team's AI platform toolchain and for contributing directly to architecture decisions on agent systems and LLM infrastructure.
- ▶ **EDA Optimization** — Reduced event-driven pipeline processing from 15–20 hours to under 15 minutes across a 50+ microservice production system.
- ▶ **IEEE Conference** — Published and presented original research on Quantum Computing at a peer-reviewed IEEE conference.

## LANGUAGES

English · Kannada · Hindi